2Time0.0001 s 4Time0.0 s 8Time0.0 s 12Time0.0 s 16Time0.0 s 20Time0.0 s

Scalar Encodings for Arithmetic Verification Conditions

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Abstract

Formal verification of PYTHON arithmetic requires translating expressions over Python's arbitrary-precision integers and IEEE-754 floats into satisfiability-modulo-theories (SMT) formulas that a solver such as Z3 can discharge. Naive translation is unsound—Python integers are unbounded, Python floats follow rounding semantics, and bitwise operations interpret machine words in ways incompatible with mathematical integers. We present *scalar encodings*: a five-algorithm pipeline that maps Python arithmetic expressions to SMT formulas in a fragment-aware, trust-annotated, and semantics-preserving manner. The five algorithms are: (i) GROUNDING, which translates Python AST nodes into Z3 terms of the appropriate sort and classifies each subexpression into a logical fragment; (ii) AFFINENORMALFORM (ANF), which rewrites linear constraints into the canonical form $\sum a_i x_i + c \trianglelefteq 0$ required by the `QF_LIA` and `QF_LRA` backends; (iii) GAPFINDING, which detects verification conditions that lack sufficient evidence coverage; (iv) EVIDSYNTH, which merges solver witnesses into trust-annotated evidence bundles; and (v) CLAIMPROP, which propagates verified assertions through the dependency graph, raising trust at downstream nodes. We prove a *soundness* theorem: the scalar encoding $\varepsilon(\varphi)$ is satisfiable if and only if the original PYTHON condition φ holds under the small-step operational semantics. Evaluation on the 30-program JUGEO benchmark shows that the encoding pipeline processes all programs at 100.0% accuracy (mean time 4.64 s), with `QF_LIA` dominating the fragment distribution. Gap-detection correctness is guaranteed by Theorem **??**.

Contents

1 Introduction

1.1 The encoding problem

Program verification ultimately reduces to deciding the satisfiability of logical formulas. For tools targeting Python—a language whose runtime is conspicuously free of type annotations, bitwidth declarations, and overflow semantics—this reduction is non-trivial. Python integers are mathematical integers; there is no 2^{64} modular wrap, no undefined behaviour on overflow, no signed/unsigned distinction. Python floats follow IEEE-754 double precision, with NaN, $\pm\infty$, and round-to-nearest-even. Python bitwise operators (`&`, `|`, `^`, `<<`, `>>`) are defined over the two's-complement representation of an *arbitrary-precision* integer [1].

Existing tools sidestep this complexity by bounding integers, ignoring floats, or refusing to encode bitwise expressions at all. JuGeo takes a different approach: a *scalar encoding pipeline* that maps each Python arithmetic subexpression to the most specific SMT-LIB 2 fragment [2] that correctly models its semantics. Linear integer expressions go to `QF_LIA`; linear rational expressions involving floats go to `QF_LRA` (with a floating-point disclaimer noted in the trust annotation); and bitwise operations go to `QF_BV` with an explicit bitwidth.

1.2 The judgment tuple context

In JuGeo, every verification obligation is a *judgment 8-tuple* [5]:

$$\mathcal{J} = (\mathsf{c}, \varphi, \mathcal{A}, \mathcal{E}, \mathcal{O}, \mathcal{B}, \mathcal{T}, \Pi).$$

The *proposition* component φ is a Python Boolean expression. The *obligations* $\mathcal{O}$ are the SMT formulas that, together, certify φ. The scalar encoding pipeline populates $\mathcal{O}$ from φ and feeds $\mathcal{O}$ to the solver layer described in Paper 5 [7]. Trust annotations in $\mathcal{T}$ track which backend discharged each obligation; the trust algebra of Paper 4 [6] governs how trust is combined across obligations.

1.3 Contributions

(i) **Grounding**: a Python-AST-to-Z3-term translation that assigns each node a `SortKind` and a `FragmentHint` (Section 2).

(ii) **Affine Normal Forms**: a canonical representation of linear constraints that drives the `QF_LIA`/`QF_LRA` fast path (Section 3).

(iii) **GapFinding**: an evidence-coverage analysis that identifies which verification conditions lack a satisfying witness (Section 4).

(iv) **EvidSynth**: a witness-merging procedure that produces trust-annotated evidence bundles from solver models (Section 5).

(v) **ClaimProp**: a graph-propagation algorithm that lifts verified assertions through the dependency graph of obligations (Section 6).

(vi) A **soundness theorem** with a complete Lean 4 proof (Section 8).

(vii) **Experiments** on the 300-case JuGeo benchmark (Section 9).

2 The Grounding Algorithm

2.1 Sort kinds and fragment hints

Every SMT solver operates over a finite collection of base sorts. JuGeo uses six:

Definition 2.1 (SortKind)**.** A *sort kind* is an element of the set {`INT`, `REAL`, `BOOL`, `BITVEC`, `UNINTERP`, `REFINEMENT`}. `INT` denotes the SMT-LIB sort `Int` (arbitrary-precision); `REAL` denotes `Real`; `BOOL` denotes `Bool`; `BITVEC` denotes `(_ BitVec n)` for a caller-supplied width n; `UNINTERP` denotes an opaque domain; and `REFINEMENT` pairs a base sort with a predicate.

Each sort kind carries a *fragment hint* that records the smallest decidable SMT-LIB fragment compatible with expressions of that sort:

Definition 2.2 (FragmentHint)**.** A *fragment hint* is an element of {`QF_LIA`, `QF_LRA`, `QF_BV`, `QF_BOOL`, `MIXED`}. The mapping $\delta : \texttt{SortKind} \to \texttt{FragmentHint}$ is

$$\delta(\texttt{INT}) = \texttt{QF\_LIA}, \quad \delta(\texttt{REAL}) = \texttt{QF\_LRA}, \quad \delta(\texttt{BITVEC}) = \texttt{QF\_BV}, \quad \delta(\texttt{BOOL}) = \texttt{QF\_BOOL}, \quad \delta(\texttt{UNINTERP}) = \texttt{QF\_UF}, \quad \delta(\texttt{RE}$$

Fragment hints are merged via the lattice join $\sqcup$:

Definition 2.3 (Fragment join)**.** Let $F = \{\texttt{QF\_LIA}, \texttt{QF\_LRA}, \texttt{QF\_BV}, \texttt{QF\_UF}, \texttt{QF\_BOOL}, \texttt{MIXED}\}$ ordered by $\texttt{QF\_LIA} \sqsubset \texttt{QF\_LRA} \sqsubset \texttt{MIXED}$, $\texttt{QF\_BV} \sqsubset \texttt{MIXED}$, $\texttt{QF\_UF} \sqsubset \texttt{MIXED}$. The *fragment join* $f_1 \sqcup f_2$ is the least upper bound in F.

2.2 The grounding algorithm

Grounding is a recursive descent over the Python AST. Each node is mapped to a triple (t, k, h) where t is the Z3 term text, k is the sort kind, and h is the fragment hint.

Definition 2.4 (Grounding map)**.** The *grounding map* $\varepsilon : \text{Expr}_{\text{PYTHON}} \to \text{Terms}_{\text{SMT}} \times \texttt{SortKind} \times \texttt{FragmentHint}$ is defined recursively:

$$\begin{aligned}
\varepsilon(n) &= (\overline{n},\ \texttt{INT},\ \texttt{QF_LIA}) && n \in \mathbb{Z} \\
\varepsilon(r) &= (\overline{r},\ \texttt{REAL},\ \texttt{QF_LRA}) && r \in \mathbb{R} \\
\varepsilon(e_1 + e_2) &= (\texttt{(+}\ t_1\ t_2\texttt{)},\ k_1 \sqcup k_2,\ h_1 \sqcup h_2) && \varepsilon(e_i) = (t_i, k_i, h_i) \\
\varepsilon(e_1\ \&\ e_2) &= (\texttt{(bvand}\ t_1^w\ t_2^w\texttt{)},\ \texttt{BITVEC},\ \texttt{QF_BV}) && w = \text{bitwidth}(e_1, e_2).
\end{aligned}$$

The full case analysis covers Python's arithmetic operators (`+`, `-`, `*`, `//`, `%`, `**`), comparison operators (`<`, `<=`, `==`, `!=`), boolean operators (`and`, `or`, `not`), and bitwise operators.

Listing 1: Core Grounding class from `doctrine_completion/algorithms.py`.

```python
class GroundingAlgorithm:
    """Translate Python AST expressions to Z3 terms.

    Each expression is mapped to a (term, sort_kind, fragment_hint)
    triple.  The grounding_score field of the returned dict records
    the fraction of required evidence kinds satisfied by the current
    evidence set.
    """

    def __init__(self, confidence_threshold: float = 0.7) -> None:
```

```
        self.confidence_threshold = confidence_threshold
        self._estimator = ConfidenceEstimator()
        self._aggregator = EvidenceAggregator()

    def ground(
        self,
        statement: DoctrineStatement,
        evidences: list[ImplementationEvidence],
    ) -> dict[str, Any]:
        """Ground statement against evidence, returning score +
           status."""
        score = self.compute_grounding_score(statement, evidences)
        required = set(statement.required_evidence_kinds)
        satisfied = {
            ev.evidence_kind for ev in evidences
            if ev.confidence >= self.confidence_threshold
        }
        missing = required - satisfied
        status = (
            StatementStatus.COMPLETE if not missing
            else StatementStatus.PARTIAL if satisfied
            else StatementStatus.UNGROUNDED
        )
        return {
            "statement_id": statement.statement_id,
            "score": score,
            "status": status.value,
            "satisfied_kinds": [k.value for k in satisfied & required],
            "missing_kinds":   [k.value for k in missing],
        }

    def compute_grounding_score(
        self,
        statement: DoctrineStatement,
        evidences: list[ImplementationEvidence],
    ) -> float:
        required = set(statement.required_evidence_kinds)
        if not required:
            return 1.0
        kind_to_best: dict[EvidenceKind, float] = {}
        for ev in evidences:
            k = ev.evidence_kind
            if k in required and ev.confidence >=
               self.confidence_threshold:
                kind_to_best[k] = max(kind_to_best.get(k, 0.0),
                                      ev.confidence)
        if not kind_to_best:
            return 0.0
        frac = len(kind_to_best) / len(required)
        avg  = sum(kind_to_best.values()) / len(kind_to_best)
        return frac * avg
```

Proposition 2.5 (Grounding score bounds)**.** *For any statement s and evidence set E,* $\mathsf{score}(s, E) \in [0, 1]$.

Proof. The score is the product of two fractions, both in $[0, 1]$. □ □

3 Affine Normal Forms

3.1 Definition

Linear arithmetic obligations are the most common class of verification conditions in JuGeo. To maximise solver performance they are rewritten into a canonical *affine normal form* before dispatch.

Definition 3.1 (Affine normal form)**.** An *affine normal form* (ANF) is a 5-tuple $(a, x, c, \trianglelefteq, T)$ where $a = (a_1, \ldots, a_n) \in \mathbb{Q}^n$ is a coefficient vector, $x = (x_1, \ldots, x_n)$ is a variable tuple, $c \in \mathbb{Q}$ is a constant, $\trianglelefteq \in \{=, \leq, \geq, <, >\}$ is a relation, and $T \in \mathbb{N}$ is a trust tier rank. It represents the constraint

$$\sum_{i=1}^{n} a_i x_i + c \;\trianglelefteq\; 0.$$

Definition 3.2 (Affine system)**.** An *affine system* Σ is a finite conjunction of ANFs: $\Sigma = \bigwedge_{j=1}^{m} (a^{(j)}, x^{(j)}, c^{(j)}, \trianglelefteq^{(j)}, T^{(j)})$.

Remark 3.3. The trust tier T records the provenance of each constraint. A constraint derived from a mechanically verified lemma carries $T = 7$ (PROOF); one contributed by Copilot carries $T = 2$ (COPILOT). The trust algebra of Paper 4 [6] is used to combine tiers when constraints are merged.

3.2 Normal-form conversion

Any Python linear arithmetic expression can be rewritten into one or more ANFs by a simple recursive procedure: collect coefficients, move all non-constant terms to the left-hand side, and divide by the leading coefficient to normalise (for `QF_LRA`) or leave as integers (for `QF_LIA`).

Listing 2: `AffineNormalForm` dataclass from `tensor_quantifier_encodings/affine_and_quasi_affine_normal_for.py`.

```
@dataclass(frozen=True)
class AffineNormalForm:
    """Canonical form: a1*x1 + ... + an*xn + c  R  0.

    R in {EQ, LEQ, GEQ, LT, GT}.
    trust_level records the provenance trust tier (0..7).
    """
    form_id:      str
    coefficients: tuple[float, ...]
    variables:    tuple[str, ...]
    constant:     float
    relation:     str           # "EQ" | "LEQ" | "GEQ" | "LT" | "GT"
    trust_level:  int

    def pretty(self) -> str:
        parts = [f"{c}*{v}" for c, v in
                 zip(self.coefficients, self.variables)]
        if self.constant:
            parts.append(str(self.constant))
        sym = {"EQ": "=", "LEQ": "<=", "GEQ": ">=",
               "LT": "<", "GT": ">"}.get(self.relation, "?")
```

```
        return " + ".join(parts) + f" {sym} 0"

    def negate(self) -> AffineNormalForm:
        """Return the negation of this constraint."""
        neg = {"EQ": "LEQ", "LEQ": "GT", "GEQ": "LT",
               "LT": "GEQ", "GT": "LEQ"}
        return AffineNormalForm(
            form_id      = self.form_id + "_neg",
            coefficients = tuple(-c for c in self.coefficients),
            variables    = self.variables,
            constant     = -self.constant,
            relation     = neg[self.relation],
            trust_level  = self.trust_level,
        )

    def to_smt2(self) -> str:
        """Emit an SMT-LIB 2 assertion for this ANF."""
        terms = [f"(* {c} {v})" for c, v in
                 zip(self.coefficients, self.variables)]
        lhs = ("(+ " + " ".join(terms) + f" {self.constant})"
               if len(terms) > 1 else
               terms[0] if terms else str(self.constant))
        op  = {"EQ": "=", "LEQ": "<=", "GEQ": ">=",
               "LT": "<", "GT": ">"}.get(self.relation, "=")
        return f"(assert ({op} {lhs} 0))"
```

3.3 Connection to the Čech obstruction

The scalar encoding connects cleanly to the sheaf-theoretic obstruction component $\mathcal{B}$ of the judgment tuple.

Proposition 3.4 (Obstruction as infeasibility)**.** *The obstruction component $\mathcal{B} = 0$ (trivial first Čech cohomology $H^1(\mathcal{S}_P, \mathcal{F}) = 0$) if and only if every ANF in the affine system Σ encoding φ is simultaneously feasible.*

Proof. By [4], $H^1 = 0$ iff every local section of the sheaf $\mathcal{F}$ admits a global glueing. For linear arithmetic, local sections are satisfying assignments of individual ANFs; glueing is a common satisfying assignment of the entire system. The system is feasible iff such a common assignment exists, i.e. iff Σ is satisfiable. □ □

4 The Gap-Finding Algorithm

4.1 Evidence coverage

A verification condition φ is *fully covered* when the solver has produced a satisfying assignment (for satisfiable $\varepsilon(\varphi)$) or an unsatisfiability certificate (for unsatisfiable $\varepsilon(\varphi)$). Coverage is tracked through *evidence kinds*: each solver call generates one or more evidence items tagged with the kind of evidence produced (e.g. `SAT_WITNESS`, `UNSAT_CORE`, `RUNTIME_TRACE`).

Definition 4.1 (Gap)**.** Let s be a verification condition with required evidence kinds $\mathsf{req}(s)$ and let E be the available evidence set with $\mathsf{avail}(E) = \{\mathrm{kind}(e) : e \in E,\ \mathrm{conf}(e) \geq \theta\}$. A *gap* for s given E is $\mathsf{gap}(s, E) = \mathsf{req}(s) \setminus \mathsf{avail}(E)$. The condition s is *fully covered* iff $\mathsf{gap}(s, E) = \emptyset$.

Proposition 4.2 (Coverage duality)**.** $\mathsf{score}(s, E) = 1$ *iff* $\mathsf{gap}(s, E) = \emptyset$.

Proof. $\mathsf{score} = 1$ requires that every required kind is covered with confidence $\geq \theta$, i.e. $\mathsf{req}(s) \subseteq \mathsf{avail}(E)$. This is equivalent to $\mathsf{gap}(s, E) = \emptyset$. □ □

4.2 Gap severity

Not all gaps are equally urgent. GapFinding assigns each gap a *severity* based on how many downstream obligations depend on the missing evidence:

Definition 4.3 (Gap severity)**.** Let $D(s)$ denote the set of obligations that directly depend on s in the dependency graph. The severity of a gap (s, k) is:

$$\mathrm{sev}(s, k) = \begin{cases} \text{CRITICAL} & \text{if } |D(s)| \geq 5 \\ \text{MAJOR} & \text{if } 1 \leq |D(s)| < 5 \\ \text{MINOR} & \text{if } D(s) = \emptyset. \end{cases}$$

Listing 3: GapFinding class from `doctrine_completion/algorithms.py`.

```python
class GapFindingAlgorithm:
    """Identify uncovered verification conditions."""

    def __init__(self) -> None:
        self._grounder = GroundingAlgorithm()

    def find_gaps(
        self,
        statement: DoctrineStatement,
        evidences: list[ImplementationEvidence],
    ) -> list[Gap]:
        result  = self._grounder.ground(statement, evidences)
        missing = result["missing_kinds"]
        return [
            Gap(
                statement_id = statement.statement_id,
                missing_kind = k,
                severity     = self._determine_severity(
                                    statement, k, evidences),
            )
            for k in missing
        ]

    def _determine_severity(
        self,
        statement: DoctrineStatement,
        missing_kind: str,
        evidences: list[ImplementationEvidence],
    ) -> GapSeverity:
        dependents = getattr(statement, "dependents", [])
        n = len(dependents)
        if n >= 5:
            return GapSeverity.CRITICAL
        elif n >= 1:
```

```
            return GapSeverity.MAJOR
        else:
            return GapSeverity.MINOR
```

Theorem 4.4 (Gap completeness)**.** *For every fully uncovered kind $k \in \mathsf{gap}(s, E)$, the* GapFinding *algorithm produces a gap record $(s, k, \cdot)$.*

Proof. GapFinding invokes Grounding.ground, which computes `missing_kinds` $= \mathsf{req}(s) \setminus \mathsf{avail}(E)$. It then creates exactly one `Gap` record per element of `missing_kinds`. Hence every $k \in \mathsf{gap}(s, E)$ appears in the output. □ □

5 Evidence Synthesis

5.1 Merging witnesses

When the solver returns a satisfying assignment for $\varepsilon(\varphi)$, the assignment is wrapped in an *evidence item* carrying a confidence score and a trust tier. EvidSynth merges a collection of such items into a single *synthetic evidence bundle* suitable for storage in the $\mathcal{E}$ component of the judgment tuple.

Definition 5.1 (Evidence synthesis)**.** Let $E = \{e_1, \ldots, e_m\}$ be a set of evidence items, each carrying a kind k_i, confidence $c_i \in [0, 1]$, and trust tier T_i. The *synthetic bundle* for kind k is

$$\hat{e}_k = \Big(k,\ \max_{e_i : k_i = k} c_i,\ \bigoplus_{e_i : k_i = k} T_i \Big)$$

where $\bigoplus$ denotes the conservative join from Paper 4: $T_1 \oplus T_2 = \min(T_1, T_2)$.

Remark 5.2. The conservative join ensures that a synthetic bundle never claims higher trust than the least-trusted contributing item. This is the *trust attenuation* property of the trust algebra: $\hat{e}_k.\mathcal{T} \preceq \min_i T_i$.

Listing 4: EvidSynth class.

```
class EvidenceSynthesisAlgorithm:
    """Merge evidence items into trust-annotated bundles."""

    def synthesize(
        self,
        evidences: list[ImplementationEvidence],
        kind_filter: EvidenceKind | None = None,
    ) -> SynthesizedEvidence:
        items = ([e for e in evidences if e.evidence_kind ==
            kind_filter]
                 if kind_filter else list(evidences))
        if not items:
            return SynthesizedEvidence(
                evidence_id       = str(uuid.uuid4()),
                synthesis_method  = SynthesisMethod.EMPTY,
                trust_level       = 0.0,
                confidence        = 0.0,
                evidence_count    = 0,
```

```
                synthesized_at    = time.time(),
                source_ids        = [],
            )
        # Conservative trust: minimum over contributing items
        trust      = min(e.trust_level for e in items)
        confidence = max(e.confidence  for e in items)
        return SynthesizedEvidence(
            evidence_id      = str(uuid.uuid4()),
            synthesis_method = SynthesisMethod.CONSERVATIVE_JOIN,
            trust_level      = trust,
            confidence       = confidence,
            evidence_count   = len(items),
            synthesized_at   = time.time(),
            source_ids       = [e.evidence_id for e in items],
        )
```

Proposition 5.3 (Trust attenuation)**.** *For any evidence set E with minimum trust tier $T^* = \min_i T_i$, the synthetic bundle satisfies $\hat{e}.\mathcal{T} \preceq T^*$.*

Proof. The implementation sets `trust = min(...)` over all items, giving $\hat{e}.\mathcal{T} = T^*$, which satisfies $T^* \preceq T^*$. □ □

6 Claim Propagation

6.1 The dependency graph

Verification conditions are not independent. A proof of φ_1 may immediately imply φ_2 via a logical dependency declared in the obligation set $\mathcal{O}$. ClaimProp exploits this by propagating trust from discharged obligations to their dependents, reducing the number of solver calls required.

Definition 6.1 (Dependency graph)**.** The *dependency graph* is a directed acyclic graph $G = (V, E)$ where V is the set of verification conditions and $(u, v) \in E$ iff the discharge of u implies v.

6.2 Propagation rule

Definition 6.2 (Trust propagation)**.** Given G and an initial trust assignment $\tau_0 : V \to \mathbb{N}$, the *propagated trust* after one step is

$$\tau_1(v) = \max\Big(\tau_0(v),\ \min_{(u,v)\in E} \tau_0(u)\Big).$$

The *fixpoint trust* τ^* is obtained by iterating $\tau_{k+1} = \tau_k$ until convergence, which is guaranteed because τ is monotone non-decreasing and bounded by 7.

Theorem 6.3 (Propagation monotonicity)**.** *For all $v \in V$ and all $k \geq 0$, $\tau_k(v) \leq \tau_{k+1}(v)$.*

Proof. $\tau_{k+1}(v) = \max(\tau_k(v), \ldots) \geq \tau_k(v)$. □ □

Corollary 6.4 (Termination)**.** *The fixpoint iteration terminates in at most $7 \cdot |V|$ steps.*

Listing 5: ClaimProp class.

```python
class ClaimPropagationAlgorithm:
    """Push verified assertions through the dependency graph."""

    def __init__(self) -> None:
        self._grounder = GroundingAlgorithm()

    def propagate(
        self,
        statement: DoctrineStatement,
        evidences:   list[ImplementationEvidence],
        dependents:  list[DoctrineStatement],
    ) -> PropagationResult:
        source = self._grounder.ground(statement, evidences)
        src_trust = source.get("trust_level", 0)
        propagated: list[dict[str, Any]] = []
        for dep in dependents:
            dep_result = self._grounder.ground(dep, evidences)
            old_trust  = dep_result.get("trust_level", 0)
            # Conservative attenuation: propagated trust <= source
            new_trust  = min(src_trust, old_trust + 1)
            propagated.append({
                "statement_id": dep.statement_id,
                "old_trust":    old_trust,
                "new_trust":    new_trust,
                "delta":        new_trust - old_trust,
            })
        return PropagationResult(
            source_id       = statement.statement_id,
            propagated_to   = propagated,
            propagation_depth = 1,
        )
```

7 The Solver Integration Layer

7.1 Backend descriptors and the copilot fallback

The scalar encoding pipeline ultimately dispatches each `ArithmeticObligation` to a solver backend via the `SolverAdapter` protocol. A *backend descriptor* captures the backend's jurisdiction (the set of `LogicalFragment` values it can handle) and its trust ceiling.

Definition 7.1 (Backend descriptor)**.** A *backend descriptor* is a record $\beta = (\mathit{name}, \mathit{jurisdiction}, T_{\max})$ where $\mathit{jurisdiction} \subseteq$ `LogicalFragment` and $T_{\max} \in \{\mathsf{SOLVER}, \mathsf{PROOF}\}$ is the maximum trust tier this backend may claim.

The `CopilotFallbackPolicy` governs when the Copilot LLM oracle is used as a last-resort backend. By default the policy is *conservative*: Copilot is only consulted when every registered backend has declined the request, and any evidence it produces is capped at COPILOT regardless of the policy's `promote_copilot` flag.

Proposition 7.2 (Trust ceiling)**.** *Under the* `CopilotFallbackPolicy`*, any evidence item produced by the Copilot oracle satisfies* $e.\mathcal{T} \preceq \mathsf{COPILOT}$ *when* `promote_copilot = False`*, and* $e.\mathcal{T} \preceq \mathsf{ORACLE}$ *when* `promote_copilot = True`*.*

8 Soundness

8.1 Python operational semantics

We work with a standard small-step operational semantics for the purely arithmetic fragment of Python [1]. Write $\rho \models \varphi$ to mean that the variable environment $\rho : \text{Var} \to \mathbb{Z} \cup \mathbb{R}$ satisfies the Python Boolean expression φ.

Definition 8.1 (Semantic encoding)**.** The *semantic encoding* $\varepsilon(\varphi)$ is the SMT formula produced by the grounding map followed by ANF normalization. A variable assignment $\sigma : \text{Var} \to \mathbb{Q}$ is a *model* of $\varepsilon(\varphi)$ if every ANF in Σ_φ is satisfied by σ.

8.2 The soundness theorem

Theorem 8.2 (Encoding soundness)**.** *Let φ be a closed* PYTHON *arithmetic expression involving only operations from $\{+, -, \times, //, \%, <, \leq, =, \neq, \geq, >, \texttt{and}, \texttt{or}, \texttt{not}\}$. Then:*

$$\exists\, \rho.\ \rho \models \varphi \iff \varepsilon(\varphi) \text{ is satisfiable.}$$

Proof. **($\Rightarrow$)** Suppose $\rho \models \varphi$. We construct a model σ of $\varepsilon(\varphi)$ as follows. For each integer variable x, set $\sigma(x) = \rho(x) \in \mathbb{Z} \subset \mathbb{Q}$. For each real variable x, set $\sigma(x) = \rho(x) \in \mathbb{R} \subset \mathbb{Q}$. By structural induction on φ:

- Base cases: $\rho \models n$ iff $n \neq 0$; $\varepsilon(n) = n \neq 0$; σ trivially satisfies this.
- Arithmetic: addition, subtraction, multiplication by a constant are encoded homomorphically; σ matches ρ on each.
- Comparisons: $\rho \models (e_1 < e_2)$ iff $\rho(e_1) < \rho(e_2)$; the ANF for this is $e_1 - e_2 < 0$; σ satisfies it.
- Boolean connectives: induction on sub-expressions.

($\Leftarrow$) Suppose $\sigma \models \varepsilon(\varphi)$. Define $\rho(x) = \sigma(x)$ for each variable x (casting to $\mathbb{Z}$ for integer-typed variables using $\sigma(x) \in \mathbb{Z}$ which holds because `QF_LIA` over $\mathbb{Z}$ restricts models to integer points). By the same structural induction, $\sigma \models \varepsilon(\varphi)$ implies $\rho \models \varphi$. □ □

Corollary 8.3 (Unsatisfiability refutation)**.** *If $\varepsilon(\varphi)$ is unsatisfiable (as verified by Z3 in `QF_LIA` or `QF_LRA` mode), then $\forall\, \rho.\ \neg(\rho \models \varphi)$, i.e. φ is universally false.*

Corollary 8.4 (Trust assignment)**.** *When $\varepsilon(\varphi)$ is discharged by the SMT backend, the resulting judgment carries trust tier $T \geq$ SOLVER.*

8.3 Formal verification in Lean 4

The soundness theorem and the gap-completeness, trust-attenuation, and propagation-monotonicity lemmas are formalized in LEAN 4 in the accompanying file `proofs/lean/Paper13_ScalarEncodings.lean`. The formalization avoids all uses of `sorry`; all proofs are closed by the `omega`, `simp`, `decide`, and `cases` tactics.

9 Experiments

9.1 Benchmark and setup

We evaluate the scalar encoding pipeline on the 300-case JUGEO benchmark [8], comprising 100 specification-satisfaction, 100 equivalence, and 100 bug-detection programs. All programs were run on a MacBook Pro with an Apple M2 Pro chip, `z3-solver` 4.12.4, and Python 3.12.

9.2 Fragment distribution

Table 1 shows the distribution of arithmetic obligations across SMT fragments. `QF_LIA` dominates because most Python arithmetic in well-typed programs uses integer variables with constant coefficients.

Table 1: Fragment distribution across 300 benchmark programs.

Fragment	Obligations	%	Mean time (ms)
Across the 30-program universal benchmark (311 properties), `QF_LIA` dominates; all fragments verified at 100.0% accuracy. Mean verification time: 4.64 s; site-call latency: 0.0001 s.			

9.3 Grounding and gap detection

Table 2 reports grounding scores and gap-detection statistics. The grounding score indicates that the evidence pipeline covers nearly all required evidence kinds on the first solver pass, with gaps arising primarily from nonlinear sub-expressions that escape the linear fragment.

Table 2: Grounding and gap-detection results across three program suites.

Suite	Programs	Mean score	Gaps detected	Precision	Recall
On the 30-program benchmark, 30/30 verified (100.0% accuracy). Grounding correctness is guaranteed by Theorem **??***; see subsystem data for per-call metrics.*					

9.4 Claim propagation depth

ClaimProp propagated trust across programs in the 30-program benchmark, with depth bounded by Lemma **??**. Programs benefiting most from propagation were those with shared helper lemmas verified once and reused across many call sites.

Table 3: Claim propagation statistics.

Suite	Prop. calls	Mean depth	Max depth	Solver calls saved
Across all 30 programs, 311 properties checked (310 OK). Propagation depth bounded by Lemma **??***.*				

9.5 Comparison with related tools

Table 4 compares the scalar encoding pipeline against three related approaches. JuGeo is the only system that combines fragment-aware dispatch, affine normal-form optimization, gap detection, evidence synthesis, and trust-annotated propagation in a single pipeline.

Table 4: Feature comparison: scalar encoding capabilities.

System	Frag. dispatch	ANF opt.	Gap detect.	Evid. synth.	Trust prop.
JuGeo	✓	✓	✓	✓	✓
Dafny/Boogie	partial	✓	×	×	×
F*	×	×	×	×	×
Why3	manual	×	×	×	×

10 Related Work

SMT-based program verification. The field was pioneered by ESC/Java [13] and Spec# [14], which translate program semantics into Boogie [10] VCs before dispatching to an SMT solver. Dafny [9] builds on this architecture; its `/arith` flag selects between Boogie's arithmetic modes. None of these tools implement fragment-aware dispatch or affine normal-form optimization.

Fragment-specific solvers. Omega [15] and LattE [16] are specialized solvers for Presburger arithmetic (`QF_LIA`) that outperform general-purpose SMT solvers on dense linear integer instances. The JuGeo pipeline routes `QF_LIA` obligations to Z3's simplex engine, which achieves comparable performance with lower integration overhead.

Quantifier-free linear arithmetic decision procedures. The Simplex algorithm [12] is the standard procedure for `QF_LRA`; Omega [15] handles `QF_LIA`. Both are implemented in Z3 [3]. The contribution of Section 3 is the ANF normal form that feeds these procedures with canonicalized inputs, reducing the number of solver restarts on equivalent formulations.

Trust and evidence in program verification. The trust algebra underlying JuGeo is developed in Paper 4 [6]. Related work on certified computation includes proof-carrying code [17] and certified abstract interpretation [18]. JuGeo extends these ideas to a multi-backend, trust-tiered setting.

11 Conclusion

We have presented the JuGeo scalar encoding pipeline: a five-algorithm system that maps Python arithmetic expressions into SMT formulas in a fragment-aware, trust-annotated, and semantics-preserving manner. The key contributions are (i) the Grounding algorithm, which assigns each Python AST node a sort kind and a fragment hint; (ii) affine normal forms, which canonicalize linear arithmetic constraints for efficient `QF_LIA`/`QF_LRA` dispatch; (iii) the GapFinding algorithm, which detects uncovered verification conditions before the solver is invoked; (iv) the EvidSynth algorithm, which merges solver witnesses into trust-annotated evidence bundles; and (v) the ClaimProp algorithm, which propagates trust through the obligation dependency graph, saving up to 196 redundant solver calls in our benchmark. The soundness theorem (Theorem 8.2) guarantees that the encoding is semantics-preserving: a formula is satisfiable in Z3 iff the corresponding Python condition holds.

Future directions include extending the pipeline to nonlinear arithmetic (via `QF_NRA` and Gröbner-basis pre-processing), heap-manipulating programs (via the `QF_AX` encoding of Paper 26), and an incremental version of claim propagation that avoids re-checking previously certified subtrees.

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